

RiboMetric Report

Analysis of `subsampled_1_percent.bam` with `gencode.v25.annotation_RiboMetric.tsv`. Completed 11:04:07 13/09/2026.

FAIL

Fails one or more Ribo-seq identity checks — frame-dependent analysis not advised without investigation. Top-line: periodicity 43.6%, CDS enrichment E = 1.35, recommended-read proportion 45.4%.

Library type elongation **Dominant read length** 30 nt **Total reads** 280,932 **Annotation** gencode.v25.annotation_RiboMetric.tsv

Tier 1 — Is this Ribo-seq?

These metrics decide whether frame-dependent analysis is defensible. Failures here mean P-site assignment and ORF calling should not proceed.

METRIC	RAW VALUE	SCORE	STATUS
Periodicity Low score: weak triplet structure; P-site assignment and ORF calling unreliable.	43.6%	43.6%	FAIL
CDS enrichment Low score: reads not enriched over coding sequence; library may reflect degradation, RNA contamination, or poor nuclease protection.	E = 1.35	25.9%	FAIL
Periodicity information Cross-check on frame dominance; large disagreement signals frame mixing or unstable offsets.	8.8%	8.8%	WARN

Tier 2 — Is it usable for my analysis?

These metrics describe whether enough usable signal remains after filtering. Weak scores are caveats, not identity failures.

METRIC	RAW VALUE	SCORE	STATUS
Usable read fraction Low score: little of the library survives recommended-read filtering for frame-sensitive work.	45.4%	45.4%	WARN
Coverage uniformity Low score: coverage dominated by a few hotspots; broad quantification may be unreliable.	91.5%	91.5%	PASS
Library saturation Low score: library under-sequenced; more reads would discover substantially more positions.	98.4%	1.6%	FAIL

Tier 3 — Technical caveats

These describe technical distortions and loss of usable depth. They colour interpretation but do not automatically fail the sample.

METRIC	RAW VALUE	SCORE	STATUS
Fragment uniqueness Low score: usable molecule diversity much lower than read depth suggests (protocol-dependent).	47.7%	52.3%	WARN
RPF unique mapping Low score: reduced confidence in locus/transcript-level quantification.	21.7%	78.3%	PASS
Alignment unique mapping Low score: many alignment rows have evidence of another reported alignment.	17.5%	82.5%	PASS
5' end integrity Low score: 5' read ends frequently clipped; offset and terminal-bias interpretation may be unreliable.	0.0%	100.0%	PASS

METRIC	RAW VALUE	SCORE	STATUS
Footprint homogeneity Low score: many transcripts have footprint-length profiles unlike the library aggregate; heterogeneous or contaminated library.	10.8%	89.2%	PASS
5' terminal evenness (KL) Low score: 5' terminal sequence bias may distort count quantification; consider correction.	0.047 bits	97.7%	PASS
3' terminal evenness (KL) Low score: 3' terminal sequence bias may distort count quantification; consider correction.	0.121 bits	93.9%	PASS
5' terminal evenness (max deviation) Low score: at least one 5' terminal dinucleotide is strongly over- or under-represented.	4.4%	95.6%	PASS

METRIC	RAW VALUE	SCORE	STATUS
3' terminal evenness (max deviation) Low score: at least one 3' terminal dinucleotide is strongly over- or under-represented.	6.1%	93.9%	PASS

Diagnostics

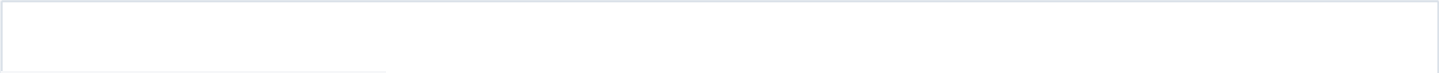
These values describe shape, context, or experiment-specific behaviour. They are shown as raw measurements — not scored — because their interpretation depends on protocol or library type.

METRIC	VALUE	NOTE
Di-some proportion	0.002	High values may indicate disome contamination or incomplete size selection.
Read-length IQR fraction (lower = tighter)	0.600	
Read-length CV (lower = tighter)	0.077	
Read-length max proportion	0.211	
Read-length bimodality coefficient (lower = unimodal)	0.047	
Start-codon enrichment ratio	1.466	
Stop-codon read-through ratio (lower = less readthrough)	0.194	

METRIC	VALUE	NOTE
5' ramp ratio	2.292	
3' drop ratio	0.876	
FLOSS median (lower = homogeneous)	0.208	

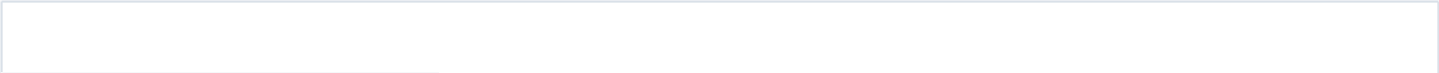
Read Frame Distribution

Frame distribution per read length



Metagene Profile

Metagene profile showing read counts by distance from the selected target. The plotted read population follows the multimap filter. MAPQ=255 unique reads; profile reads n=171,916/171,916; window reads: start n=12,219, stop n=5,360.



Metagene Heatmap

Metagene heatmap showing A-site distance from the selected target by read length. The plotted read population follows the multimap filter. MAPQ=255 unique reads; profile reads n=171,916/171,916; window reads: start n=12,219, stop n=5,360.



mRNA Reads Breakdown

Shows the proportion of the different transcript regions represented in the reads



mRNA Reads Breakdown over Read Length

Shows the proportion of the different transcript regions represented in the reads over the different read lengths.



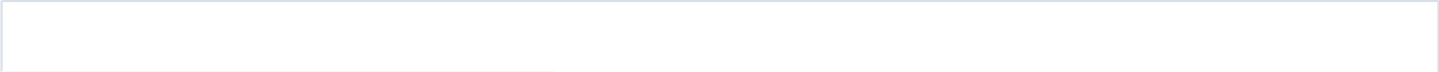
Read Frame Triangle

Triangle plot showing the distribution of read frames for the full dataset

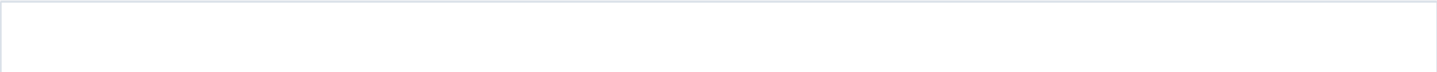


Ligation Bias Distribution

Distribution of end bases for the full dataset



Nucleotide Composition



Read Length Distribution

Distribution of read lengths for the full dataset



Recommended Read Lengths

Per-read-length 3-nt periodicity. Blue bars are recommended for downstream P-site assignment / ORF calling (periodic enough and carrying enough reads); grey bars are not.



Gene-body Coverage

A-site density across relative CDS position. A 5' bump is the translation ramp; a 3' decline indicates drop-off. A flat line at 1.0 is perfectly uniform elongation.



Library Complexity

Expected distinct A-site positions recovered vs sequencing depth. A curve that has flattened by full depth is saturated (extra sequencing adds little); a still-rising curve is under-sequenced.



FLOSS Heterogeneity

Distribution of per-transcript FLOSS scores (footprint-length departure from the library aggregate). A long right tail means many transcripts have anomalous length profiles — a heterogeneous or contaminated library.

